

C404A1	Runtime tries to receive data too often	Communication overload	Same as C404A0
C450A	Force-Torque sensor		
C450A0	Sensor data invalid	Force-Torque sensor is defective or not mounted correctly	a) Do a Complete rebooting sequence b) Check for loose connections c) Update software d) Contact your local Universal Robots service provider for assistance.
C450A1	Sensor cannot be used, therefore it is disabled	Force-Torque sensor version is newer than the Robot software	Same as C450A0
C450A2	Force-Torque sensor	Channel signal became invalid	
C450A3	Force-Torque sensor	Check frequency	
C450A4	Force-Torque sensor is expected, but it cannot be detected	Force-Torque sensor is expected, but no signals from the sensor can be detected.	a) Do a Complete rebooting sequence b) Check for loose connections c) Check for damages to the Tool/sensor d) Update software e) Contact your local Universal Robots service provider for assistance.
C450A5	Force-Torque sensor is detected but not calibrated	Force-Torque sensor is installed, but no calibration was found	Same as C450A5
C499A	Motor encoder calibration status		a) Conduct a complete rebooting sequence b) Update software c) Contact your local Universal Robots service provider for assistance
C499A0	Actual position isn't final. Position error	Motor encoder calibration status	Same as C499A
C499A1	Actual position has an error. Position error	Motor encoder calibration status	Same as C499A

C499A2	Actual position has a an error and isn't final. Position error	Motor encoder calibration status	Same as C499A
C499A3	Target position	Motor encoder calibration status	Same as C499A
C499A4	Actual average position	Motor encoder calibration status	Same as C499A
C499A5	Actual position variance	Motor encoder calibration status	Same as C499A
C499A6	Actual position min to max delta	Motor encoder calibration status	Same as C499A
C499A7	Actual sample position number	Motor encoder calibration status	Same as C499A
C720A2	LVD (low voltage detection)	Low Voltage warning level reached	Same as C720A
C710A	ROM Test		a) Do a complete rebooting sequence b) Update software c) Contact your local Universal Robots service provider for assistance
C710A0	Critical error	A critical error occurred during ROM validation	Same as C710A
C710A1	ROM corrupted	A critical error occurred during ROM validation	Same as C710A
C710A2	Failed to validate CRC on invariant data in RAM	A critical error occurred during ROM validation	Same as C710A
C710A3	Unexpected size of invariant data	A critical error occurred during ROM validation	Same as C710A
C710A4	Failed to validate CRC on code segment in RAM	A critical error occurred during ROM validation	Same as C710A
C720A	Low Voltage Detection (LVD)		a) Do a Complete rebooting sequence b) Check for loose connections d) Update software e) Contact your local Universal Robots service provider for assistance

C720A0	Reset due to Low Voltage Detection (LVD) or power off	Voltage needed values is not reaching the necessary values to work. Out of range	Same as C720A
C720A1	Low Voltage warning level reached	Voltage needed values is not reaching the necessary values to work. Out of range	Same as C720A
C740A	Hardware monitoring		a) Do a Complete rebooting sequence b) Update software c) Contact your local Universal Robots service provider for assistance
C740A0	Critical error	A critical error occurred during hardware monitoring	Same as C740A
C740A1	1V2 voltage is outside of the allowed range		Same as C740A
C740A2	1V8 voltage is outside of the allowed range		Same as C740A
C740A3	2V5 voltage is outside of the allowed range		Same as C740A
C740A4	3V3 voltage is outside of the allowed range		Same as C740A
C740A5	PC 3V3 voltage is outside of the allowed range		Same as C740A
C740A6	uA's 3V3voltage is outside of the allowed range		Same as C740A
C740A7	uB's 3V3 voltage is outside of the allowed range		Same as C740A
C740A8	5V voltage is outside of the allowed range		Same as C740A
C740A9	12V voltage is outside of the allowed range		Same as C740A

C740A10	24V voltage is outside of the allowed range		Same as C740A
C740A11	48V voltage is outside of the allowed range		Same as C740A
C740A12	58V voltage is outside of the allowed range		Same as C740A
C740A13	-4V voltage is outside of the allowed range		Same as C740A
C740A14	Robot voltage is outside of the allowed range		Same as C740A
C740A15	Robot current is outside of the allowed range		Same as C740A
C740A17	Inrush protected 48V voltage is outside of the allowed range		Same as C740A
C740A20	24V SPI IO voltage	24V voltage IO is outside its limits (Below 23,0V above 25,7V)	a) Do a Complete rebooting sequence b) Check for loose connections c) Check for damages to the Tool/sensor d) Update software e) Contact your local Universal Robots service provider for assistance
C740A21	24V SPI IO current	24V current IO is outside its limits	a) Do a Complete rebooting sequence b) Check for loose connections c) Check for damages to the Tool/sensor d) Update software e) Contact your local Universal Robots service provider for assistance

C740A22	The solenoid voltage is not 0 before activation		Same as C740A
C740A23	The solenoid voltage is not 48V after activation		Same as C740A
C740A24	The left Three-Position Enabling button is inconsistent	Check for connections problem. Check 3PE functionality	a) Do not press the button repeatedly b) Remove any pressure off the button c) Press the button with more pressure, if light pressing may not work d) Update software e) Contact your local Universal Robots service provider for assistance
C740A25	The right Three-Position Enabling button is inconsistent	Check for connections problem. Check 3PE functionality	Same as C740A24
C740A26	State for the Three-Position Enabling button	Check for connections problem. Check 3PE functionality	Same as C740A24
C900A	Debug message data		

7.2. LED indicators and Fuses on Safety Control Board

Safety Control Board (SCB)

1. Fuse 48 V:

The 5A fuse (F801) "48V" protects all 48V in the system inclusive of Euromap from over current.

This information is only for troubleshooting. Do NOT replace the fuse under any circumstances.

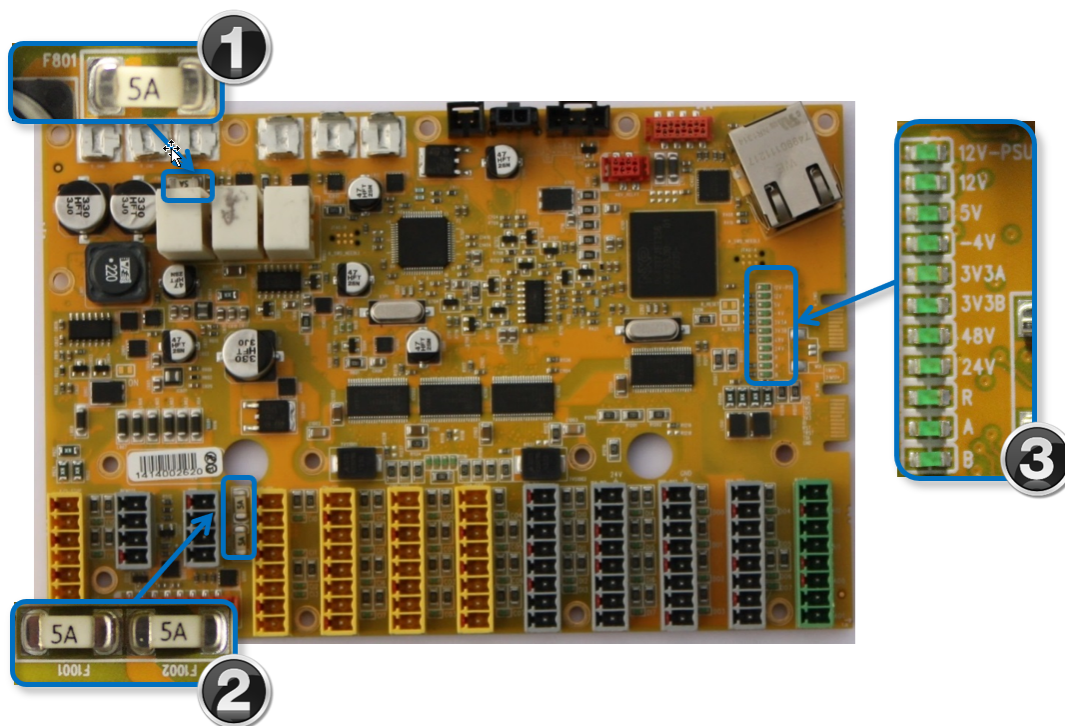
In case of failure, ONLY replace the SCB with a new tested board.

2. Fuse 24 V:

2 x 5A fuses (F1001 and F1002) in parallel protect the DI/DO 24 V supply on the Safety Control Board, whether the 24 V is from the controller or external power supply. Do NOT replace the fuses under any circumstances. In case of failure ONLY replace the SCB with a new tested board.

3. LED indicators:

- 12V-PSU On when the power plug is connected.
- 12V System: On when the main controller power has been activated.
- 5V On when "12V System" is on and indicates that 5V is present.
- -4V On when "12V System" is on and indicates that - 4V to analog I/O is present.
- 3V3A On when 5V is on and indicates 3.3V for logic Safety circuit A is present.
- 3V3B On when 5V is on and indicates 3.3V for logic Safety circuit B is present.
- 48V Indicates 48 V is present on the Safety Control Board
- 24V 48V is detected and ok, indicates that internal 24 V is present for I/Os
- R 48 V present on robot arm
- A Indicates Status for Logic A: a blink sequence
- B Indicates status for Logic B: a blink sequence



7.2.1. Normal Startup Sequence for a CB3.x UR5

1. The 12V-Power supply LED is on when the power plug is connected to a working power supply.
2. When the power button on the teach pendant is pressed, all LED indicators are turned on except for the 48V, 24V and R LEDs. The A and B LEDs also exhibit a special behavior by intermittently turning off and on ("blinking") once triggered.
3. The final phase of the startup sequence occurs (immediately) after the Polyscope software is done loading. At this stage, the 48V and 24V LED indicators become active (are switched on).

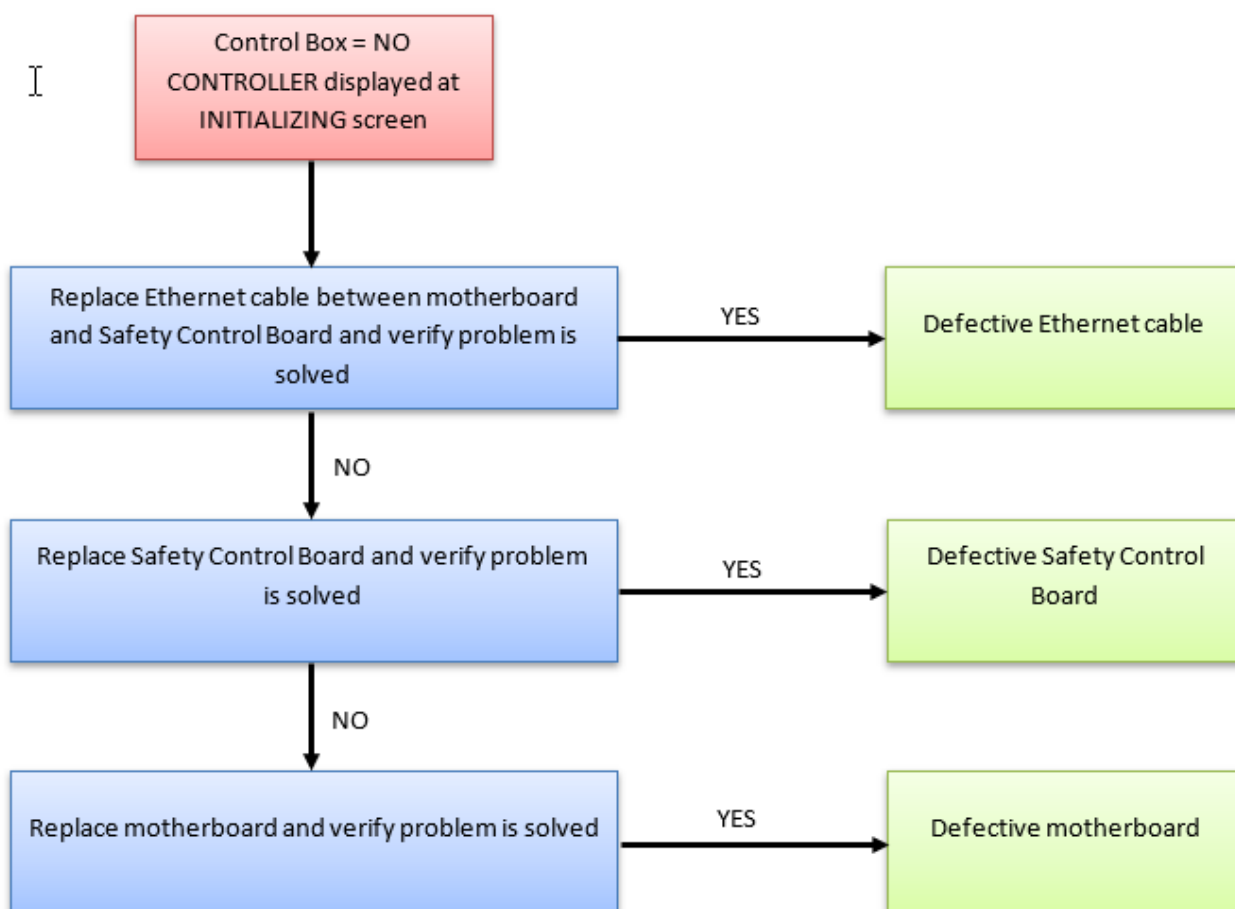
If the 48V LED indicator is off all the time in the startup sequence, measure the voltage as described below:

See the E-Plan diagram: [Schematic Overview](#)

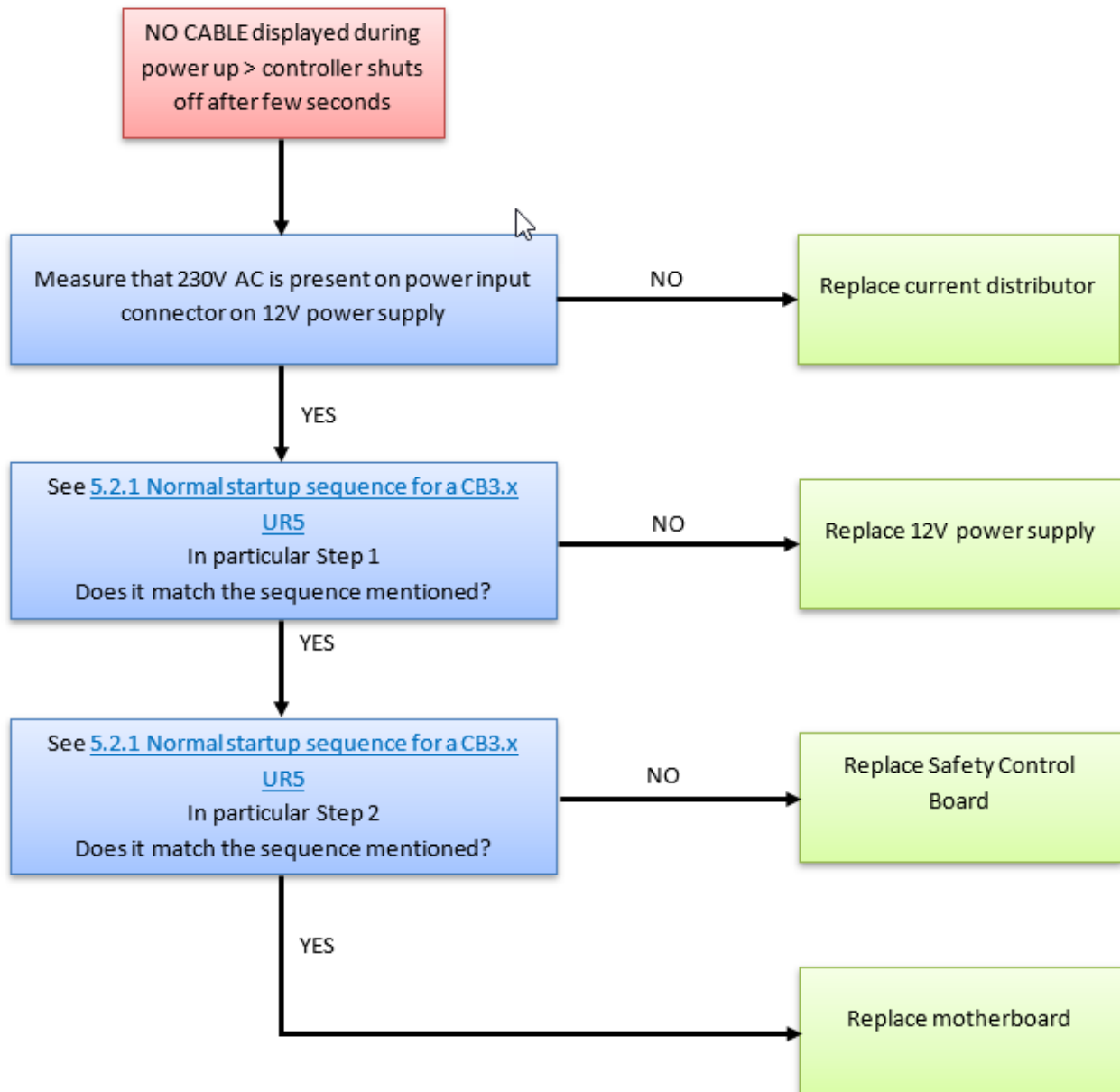
1. Measure the 48V on the Safety Control Board (SCB) where the 48V comes from the 48V power supply. And check this 1 second pulse.
 1. The voltage is measured on the Safety Control Board.
That means the Safety Control Board is defect.
 2. No voltage is measured on the Safety Control Board. Then measure the 230 V on the input side of the 48V power supply.
If the voltage pulse of 1 second is present the Power supply is defect.
 3. No voltage is measured on the input of the power supply. Then measure the 230 V on the input side of the Current distributor.
If the voltage is present the current distributor is defect.

7.3. Error Phenomena

7.3.1. Control Box: NO CONTROLLER Displayed in Initializing

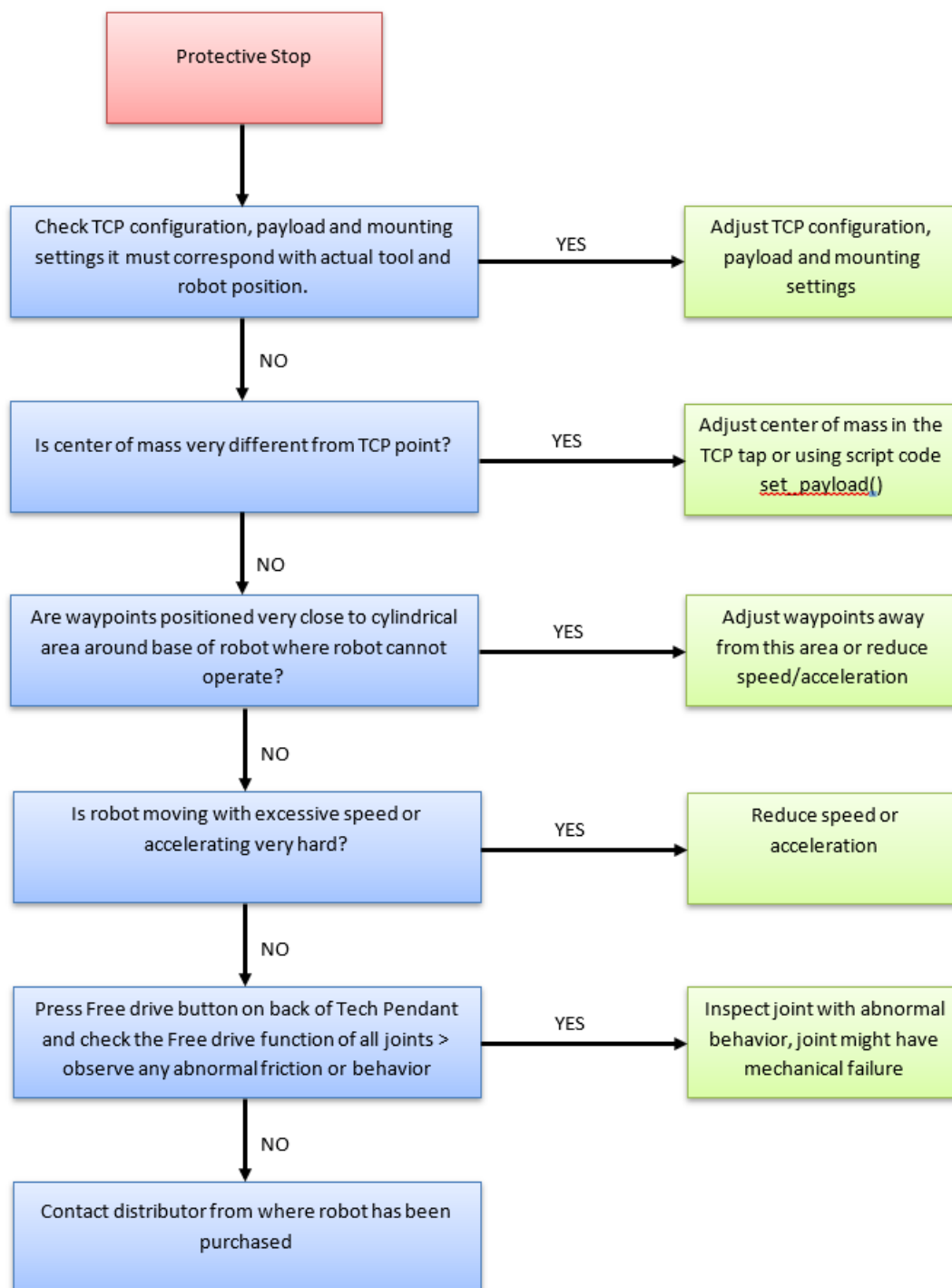


7.3.2. NO CABLE Displayed During Power Up



7.3.3. Protective Stop

Read also article 18939 on the support site www.universal-robots.com/support



Ignoring protective stops is considered abuse of the robot which voids the warranty.

These faults can be hidden from view under the following two circumstances:

1. Personnel simply resetting a fault without review of why the fault has happened.

WARNING

Ignoring protective stops masks fault detection!

Pay careful attention to protective stops.

Learn why they happen to improve your programs and preserve fault detection!

Protective stops must never be acknowledged and reset automatically, it must always be a deliberate action by a user to resume after a protective stop.

In general, protective stops are designed to inform the user that the robot has stopped due to an external event such as hitting an obstacle or similar. In case the robots are pushed close to the limits, the robots can generate protective stops to indicate that they are not capable of following the desired trajectory. After a protective stop, the following must be done before resuming operation:

i. **If there has been a collision or similar:**

Remove the obstacle and ensure that operators are out of the way before resuming operation. See Section 2 of Service Manual (see link below).

ii. **If there has been NO collision or similar:**

The robot is operating too close to the limits, the application should be adjusted to reduce the load on the robot, for instance by reducing accelerations, by correct use of blends or similar measures.

Protective stops are indications of issues, including program or production issues – not only safety issues. An application which results in daily protective stops is not designed correctly and needs to be modified.

2. If an automatic acknowledgment and reset of a protective stop has been programmed – no one will see the protective stop.

WARNING

Automatic acknowledgment and reset of protective stops masks faults that will lead to a failure condition.

When there is a protective stop, verify the cause.

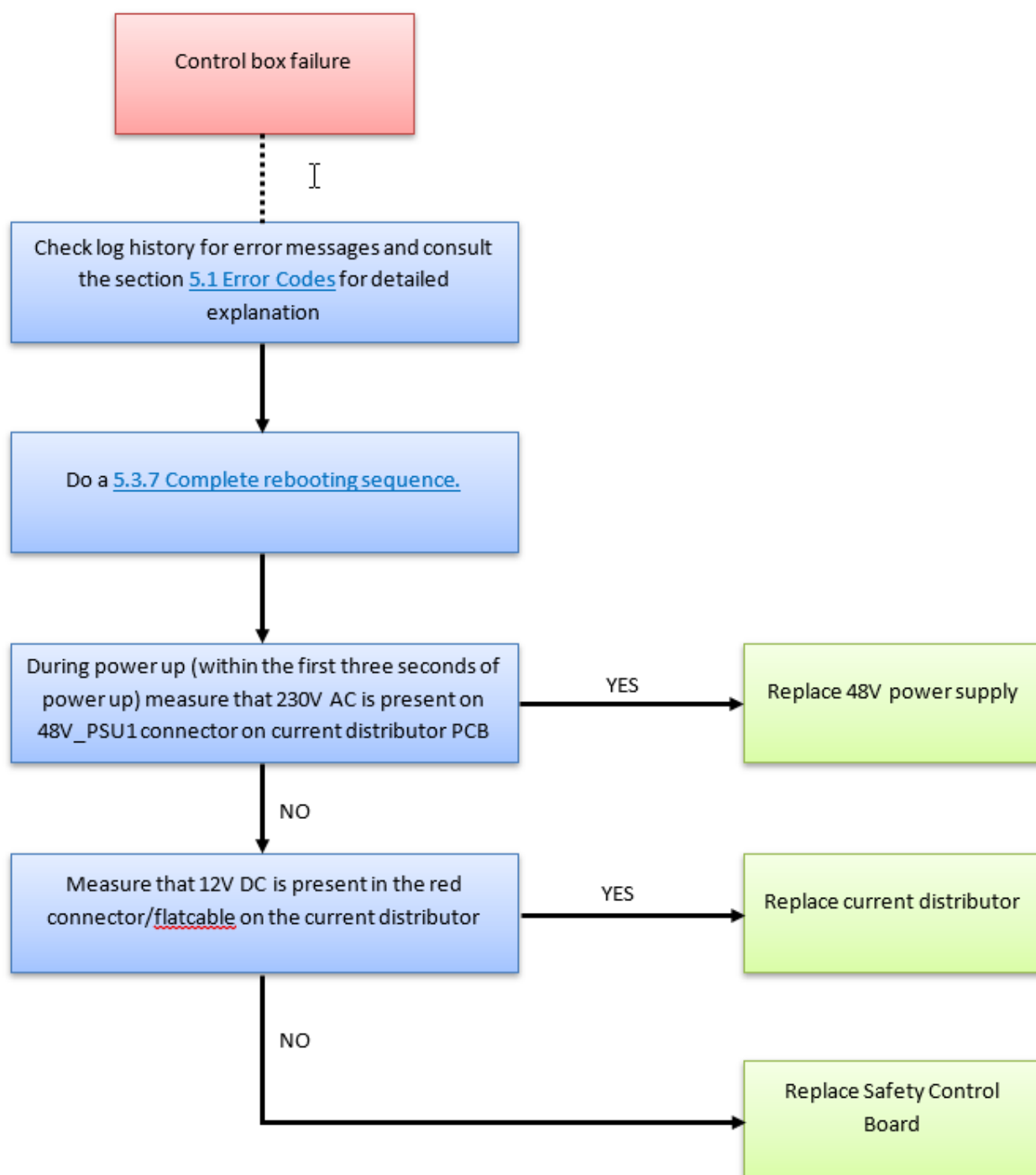
If there has been no collision, then adjust the program

If an integrator has set-up the application program to do automatic acknowledgement and reset of protective stops, the customer should contact the integrator change the program immediately, as such override voids the product warranty and masks fault detection.

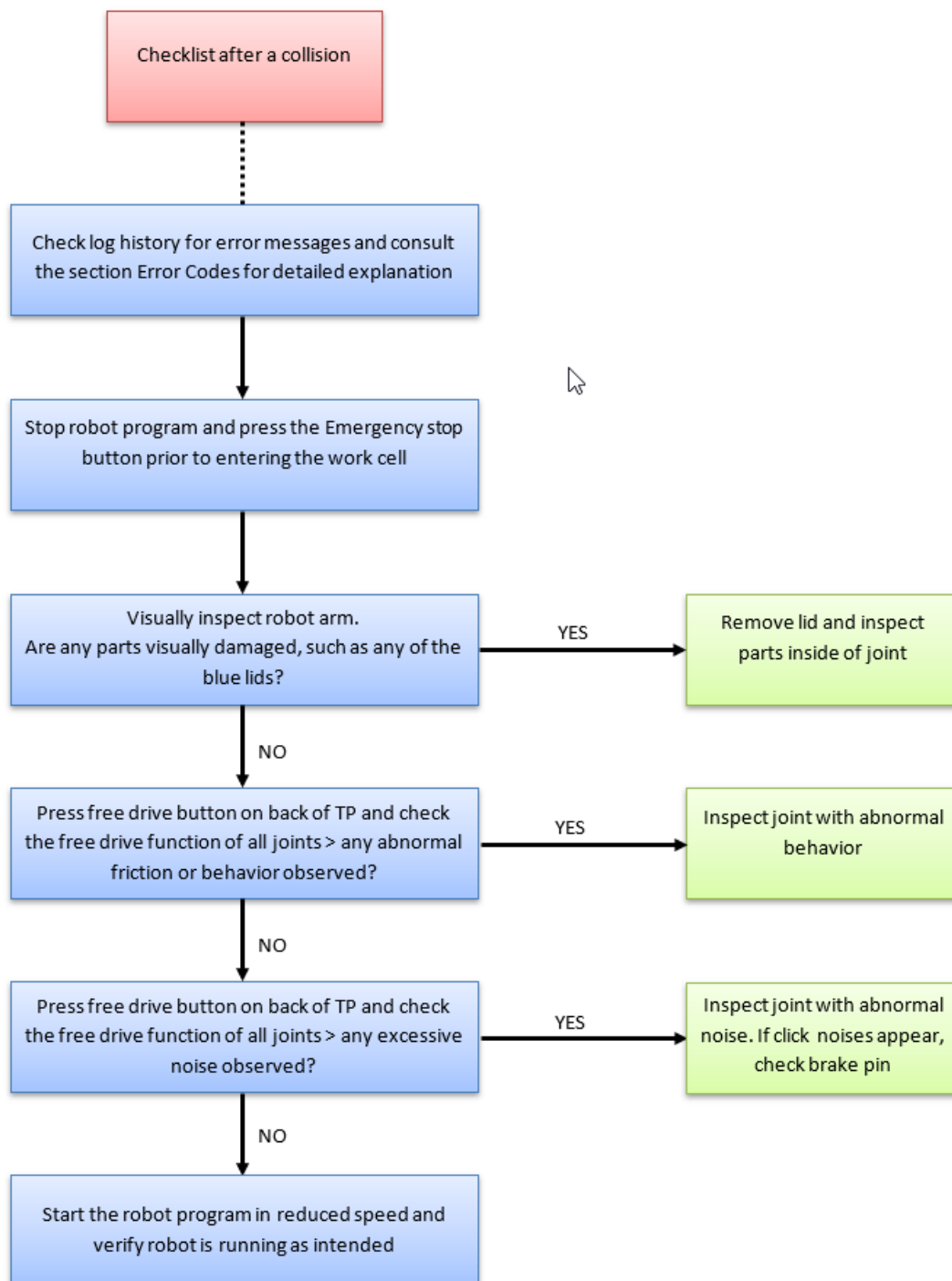
7.3.4. Power on Failure in Initializing

If power turns off a few seconds after Robot Power is turned On in the Initializing window, there are many possible causes for this phenomenon.

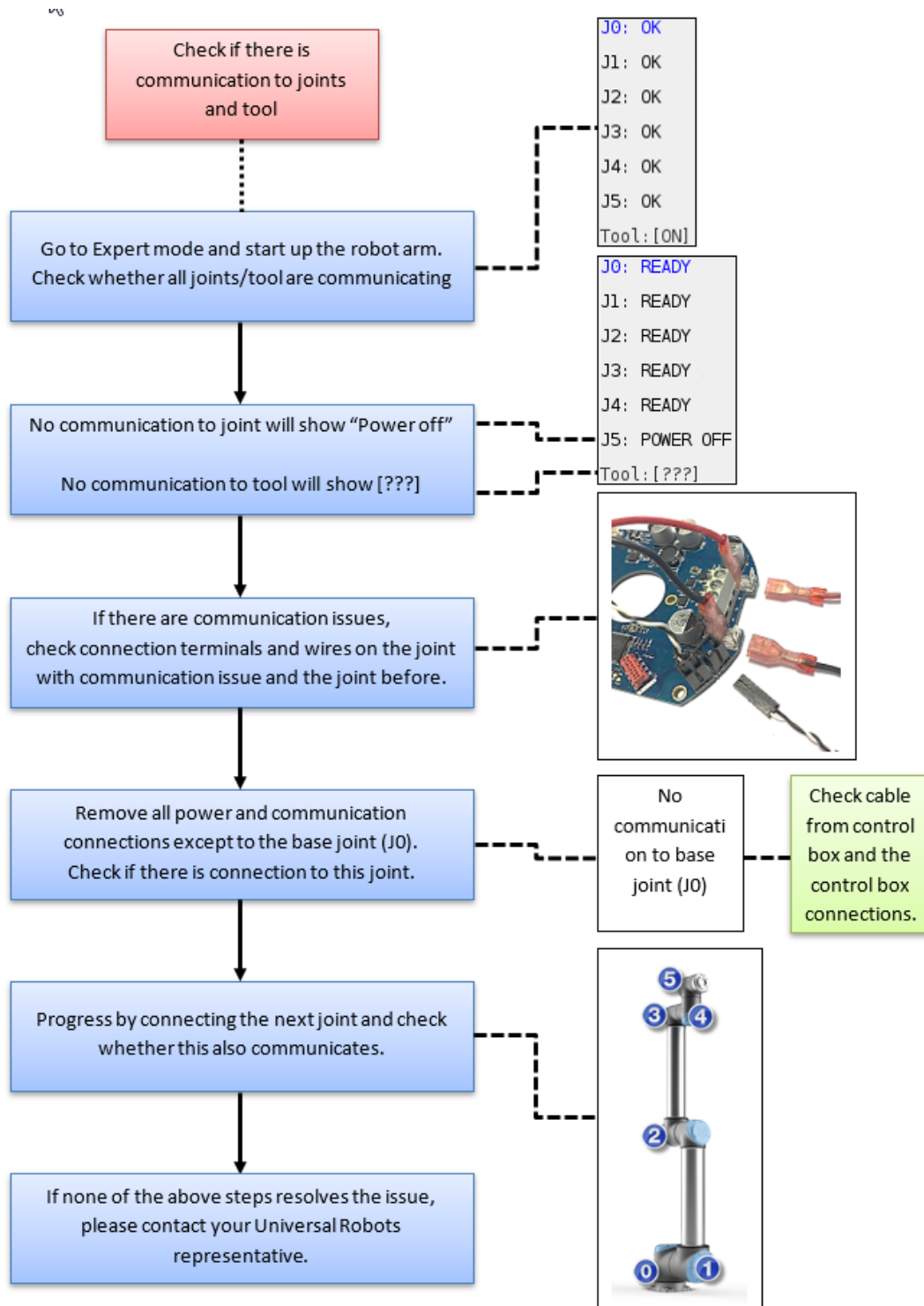
Most likely it is a control box failure or a communication failure with a joint or the tool.



7.3.5. Checklist After a Collision

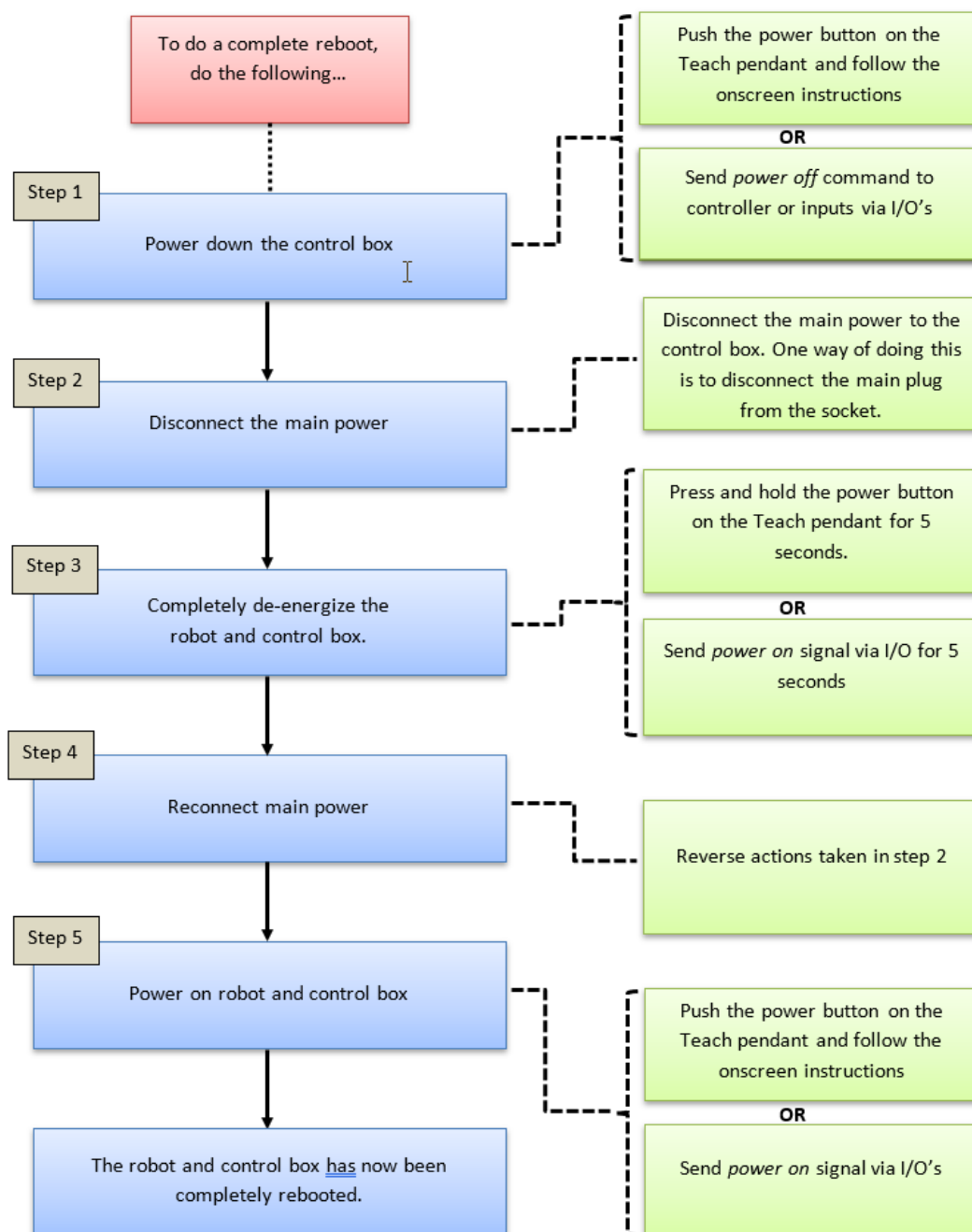


7.3.6. Communication to Joints and Tool



7.3.7. Complete Rebooting Sequence

To do a complete reboot of the robot system follow these 5 steps:



7.4. Electrical Drawing

Diagrams in pdf or in E-plan format, can be found on the support site: www.universal-robots.com/support/

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F26_001_UR

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UNIVERSAL ROBOTS

Company

Universal Robots

Drawing number

3.0.5.2

Project name

Universal Robots UR5

Created on

25-09-2014

Edit date

02-12-2015

Number of pages

19

Date

02-12-2015

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Universal Robots

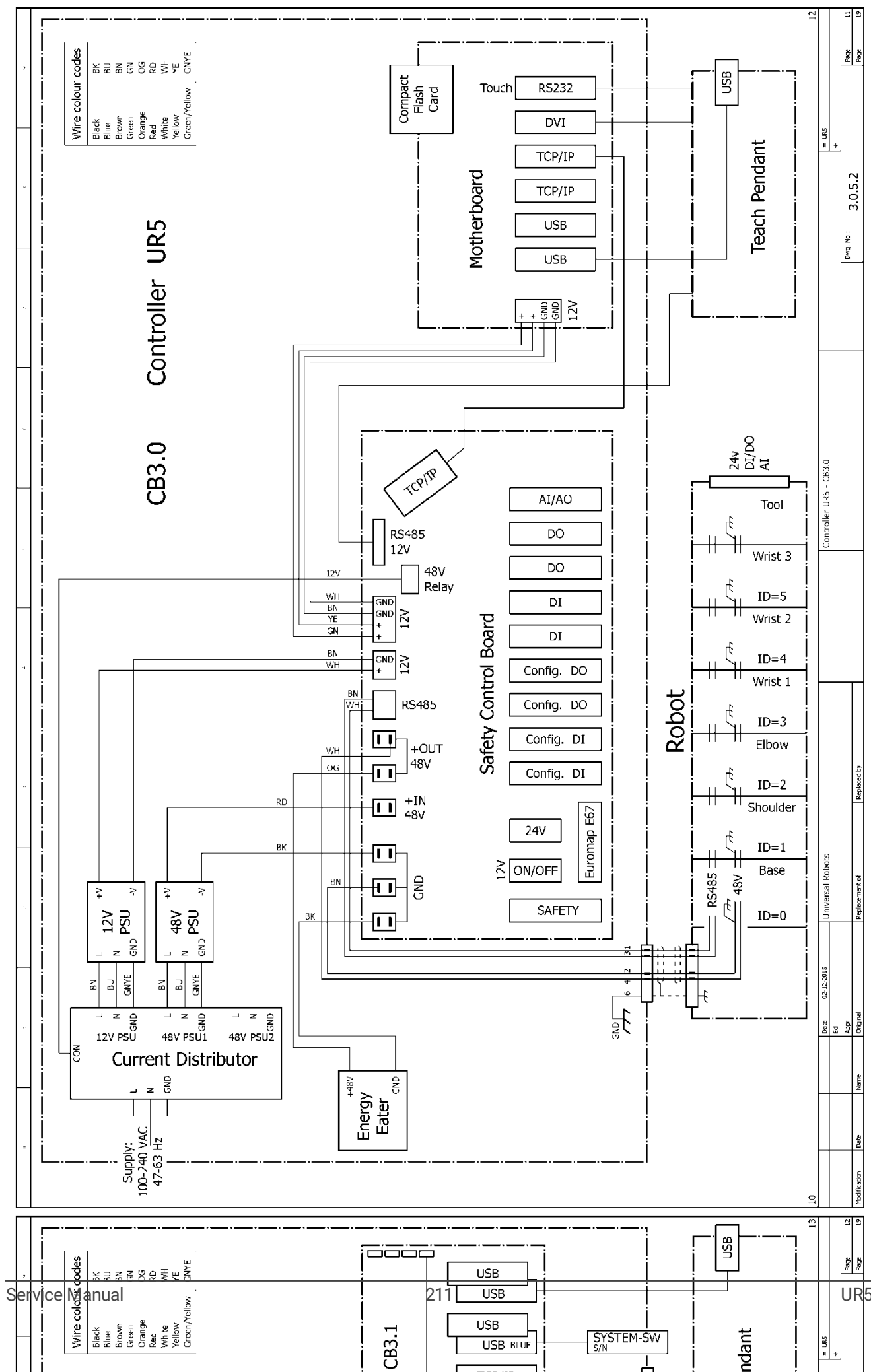
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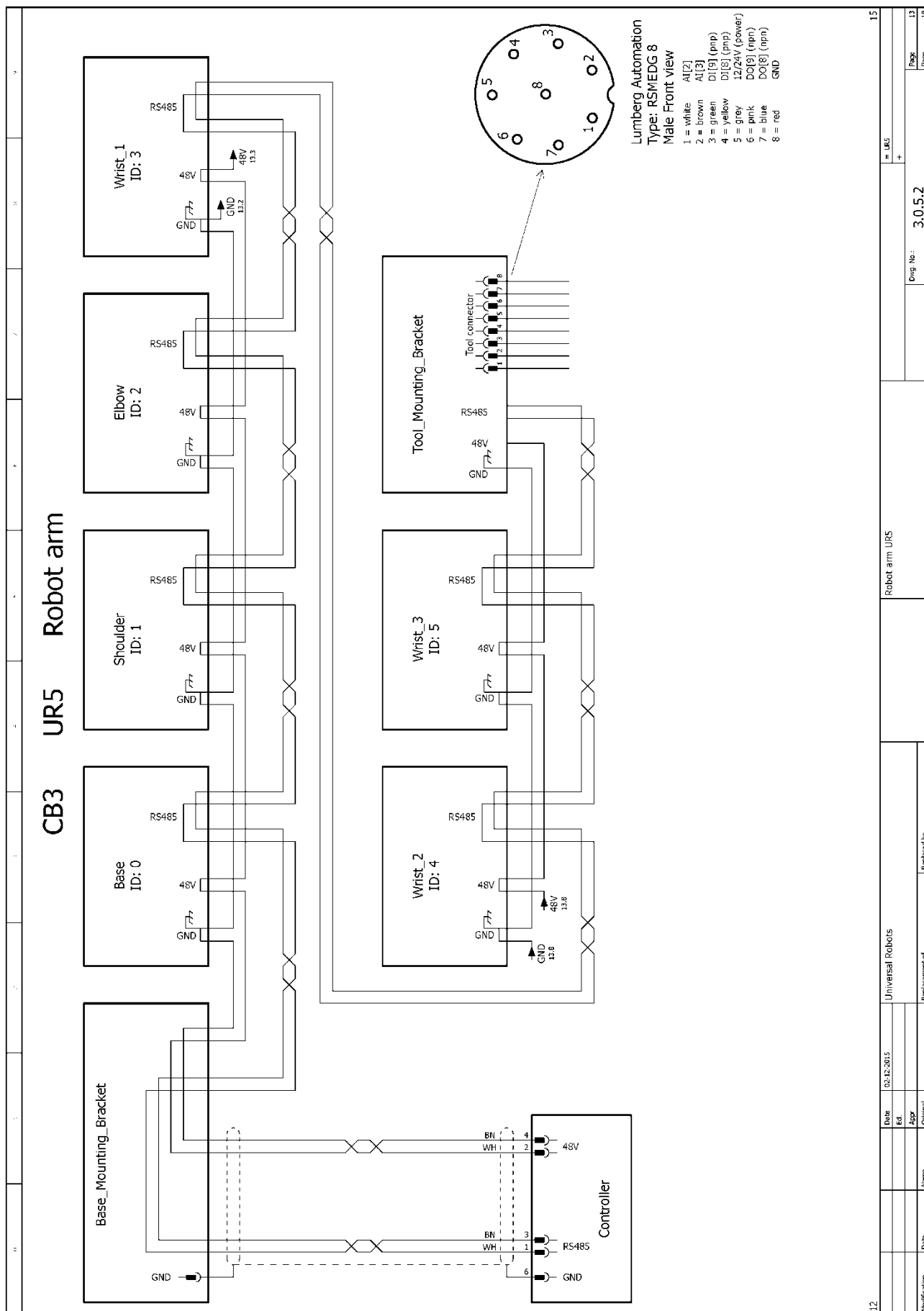
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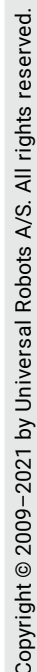
7.4.1. Schematic Overview

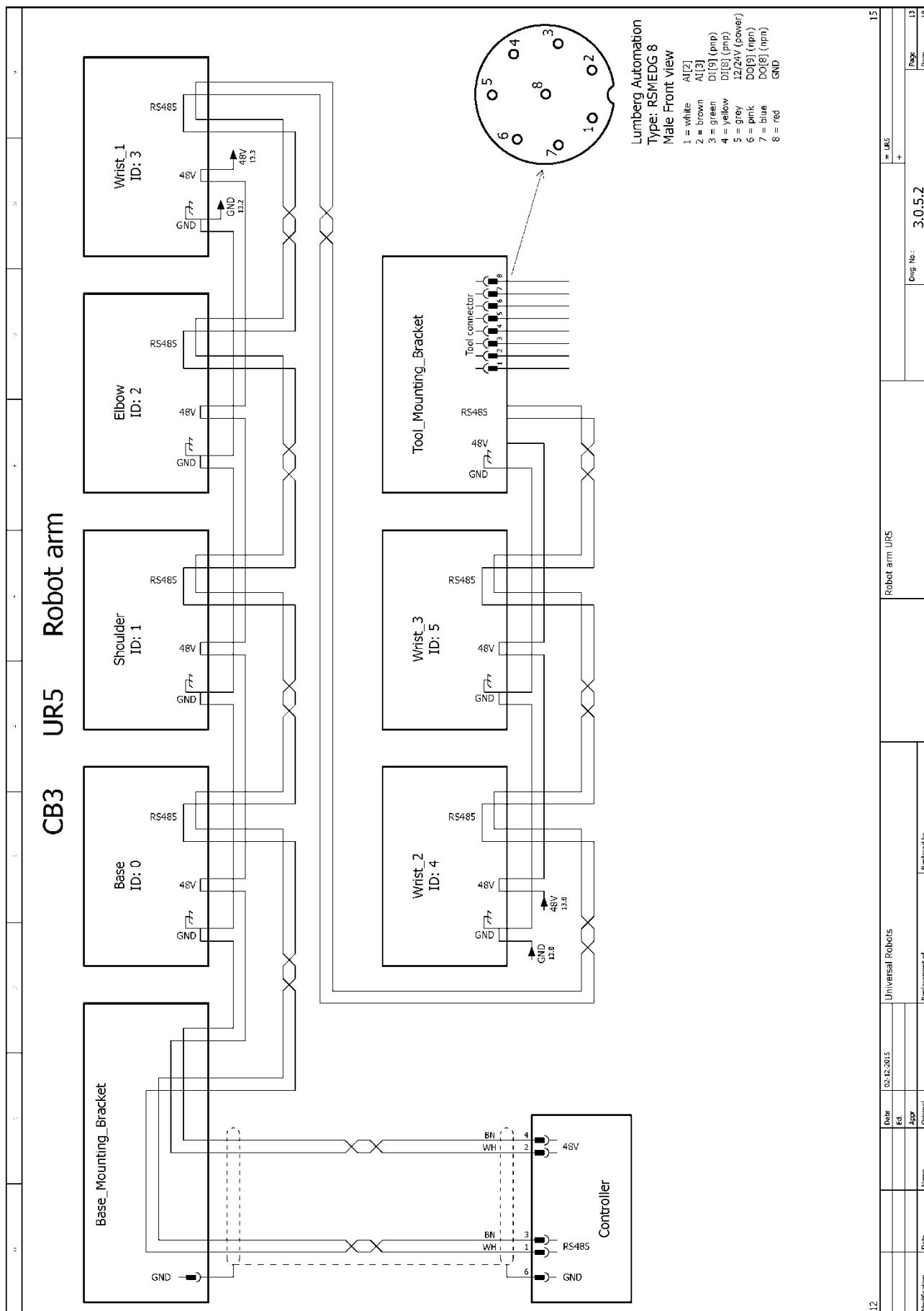
Diagrams in pdf or in E-plan format, can be found on the support site: www.universal-robots.com/support

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Drawing number		3.0.5.2							
Project name		Universal Robots UR5							
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Edit date		02-12-2015							
		Number of pages 19							
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Date		02-12-2015		Universal Robots				3.0.5.2	
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7.4.2. E-Plan Diagrams

Diagrams in pdf or in E-plan format, can be found on the support site: <https://www.universal-robots.com/support/>

